

CONFETTI: CONvolutional FibEr TracT Inference with Multi-Resolution Graphs

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Abstract. Per-tract-bundle mean summaries are commonly used for diffusion MRI based white-matter (WM) microstructure analyses to reduce the dimensionality of the data, but collapsing each tract to a set of scalars (one per diffusion metric) discards the within-tract spatial structure and vastly reduces spatial localization of effects. We introduce **CONFETTI** (**CON**volutional **FibEr TracT** Inference), a WM analysis pipeline built around three components: (i) a hierarchical multi-resolution neighborhood graph over parametrized fiber axes that retains within-tract spatial structure and between-tract proximity; (ii) an implicit-neural-representation-based imputation for missing diffusion tract profiles; and (iii) a Graph Convolutional Network (GCN) analysis including per-tract/node/covariate interpretable attribution. We evaluate CONFETTI with three structurally distinct architectures: 1) single-level GCN, 2) multi-scale parallel-branch concatenation, 3) hierarchical Graph U-Net, in a study of WM microstructure in infants with Down syndrome (DS). We investigate DS classification as well as prediction of later outcome scores. We compare CONFETTI against per-tract mean baselines (logistic regression and XGBoost). Our results show that while the classification performance is comparable between baselines and the three GCN models, the attribution results show a spatially finer pattern, as well as the emergence of the anterior commissure (AC) as a tract of higher relevance in DS. It is the top-ranked tract for the GCN-based models, while neither baseline lists it in its highly ranked tracts. Via several follow-up experiments, we show that this spatial finding is robust, and that it requires CONFETTI's within-tract spatial coupling to emerge, providing a showcase result for the proposed convolutional tract analysis with CONFETTI over traditional per-tract mean summaries.

Keywords: Graph Convolutional Network · Diffusion MRI · Fiber Tractography · Spatial Attribution · Down Syndrome.

1 Introduction

White matter (WM) fiber based microstructure analysis from diffusion MRI data typically reduces fiber bundles to one number per diffusion metric: a tract-average Fractional Anisotropy (FA), or Neurite Density Index (NDI) or similar [12]. This compression is convenient for traditional statistical analyses and classifiers [9,

8] but deliberately removes the along-tract spatial profile reducing the spatial localization of effects [11, 14]. Fiber-bundle methods such as the atlas based UNC-Utah NA-MIC DTI framework [3] or similar ones such as the Automated Fiber Quantification family [11] generate along-tract, per-node diffusion metric profiles. Current profile analysis methods focus on per-tract analyses, and no all-tract/whole-brain methods exist that operate on these 1D profiles *jointly across tracts*, treating the whole white-matter representation as a single geometric object.

CONFETTI: We address this gap with **CONFETTI** (Convolutional Fiber Tract Inference), a multi-resolution graph convolutional pipeline for WM microstructure analysis combining (1) a hierarchical neighborhood graph over parametrized fiber axes with nested levels $L_0, \dots, L_5$ shared across subjects, that captures both within-tract continuity and between-tract proximity, (2) a per-subject implicit neural representation (INR)-based imputation [10] for missing profiles, and (3) a GCN analysis stack with an out-of-fold interpretable attribution back-end (permutation importance [8], integrated gradients [7], and sparse GNNExplainer mask [6]). CONFETTI is architecture-agnostic; we compare here three structurally distinct GCN heads: a single-level GCN [1], a multi-scale parallel-branch concatenation [2, 4], and a hierarchical Graph U-Net [5].

Application: We evaluate CONFETTI in a study of WM microstructure in infants with Down syndrome (DS), a cohort that offers a methodological stress test: per-tract mean baselines already saturate the classification AUC, so any added value from CONFETTI must come from *interpretation*. We benchmark CONFETTI against logistic regression and XGBoost on per-tract DTI/NODDI means under matched nested cross-validation on (a) DS classification and (b) prospective regression of 24-month behavioral outcomes from 6-month imaging, then run focused validation experiments on the divergent finding.

Contributions: (i) **CONFETTI**, a multi-resolution graph convolutional pipeline for WM microstructure analysis with INR-based per-subject imputation and an out-of-fold GCN-based attribution stack. (ii) Three GCN instantiations (single-level, multi-scale concat, Graph U-Net) that share the same CONFETTI front-end and attribution back-end, making cross-architecture comparison meaningful. (iii) A DS showcase in which CONFETTI matches strong per-tract mean baselines in accuracy while the *anterior commissure* (AC) emerges as a top-ranked spatial finding that neither baseline lists in its highly ranked tracts; follow-up experiments establish the finding as biological signal rather than methodological artifact. (iv) A prospective-regression head showing that the same dominant spatial pattern predicts 24-month behavioral outcomes from 6-month imaging.

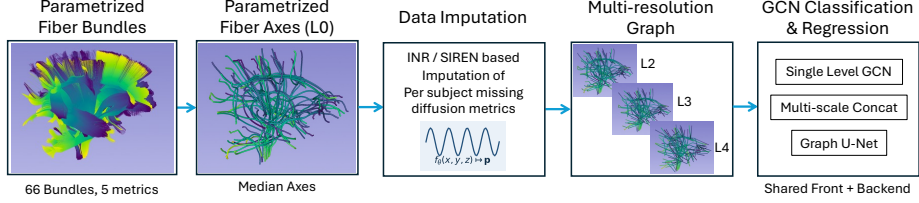


Fig. 1. CONFETTI overview. (a) Per-subject parametrized fiber bundles (66 tracts, 5 DTI/NODDI metrics); (b) per-tract parametrized fiber axes at L_0 , computed as a median location each bundle’s fibers; (c) SIREN per-subject INR imputation of missing profiles; (d) hierarchical multi-resolution neighborhood graph (L_2 , L_4 , L_5 shown); (e) three GCN heads sharing the CONFETTI front-end and attribution back-end.

2 The CONFETTI Pipeline

2.1 From fiber tracts to multi-resolution graphs

Figure 1 summarizes the pipeline; the remainder of this section details each component in turn.

For each of 66 pre-segmented white-matter tracts, we compute a 1D parametrized fiber axis by binning per-subject fiber points along their canonical arclength and taking the per-bin median, with end-clipping for outliers. A neighborhood graph is built over the concatenation of all axis points: two points are connected if they are adjacent on the same axis or within $1.5 \times$ the median nearest cross-tract distance, with edge weights $\exp(-d^2/\sigma^2)$. From the finest graph L_0 ($\sim 4,766$ nodes) we recursively subsample to produce $L_1, \dots, L_5$ while preserving each tract’s endpoints. This hierarchy is computed once and shared across subjects.

2.2 Per-subject node features and INR imputation

Fiber QC procedures commonly remove 5-10% of fiber profiles resulting in uneven profile coverage which needs correction via data imputation. At each axis node we attach five diffusion metrics (FA, RD, AD, NDI, ODI) and four spatial features (x, y, z, ℓ) for a 9-D input. Any missing profile data were imputed using an implicit neural representation (INR) model using SIREN [10] activation functions (200 epochs, $\omega_0 = 10$). Per-fold training-set statistics are used so that the imputation respects the nested-CV split. Three subject-level covariates (sex, gestational age at birth, DWI artifact level) are mean-imputed per fold and concatenated to the graph embedding before the MLP head, keeping demographics separable from the spatial pattern under attribution. We compare the INR imputation versus traditional baselines (kNN with 5 and 10 neighbors, linear, mean, median imputation).

2.3 GCN architectures for CONFETTI

CONFETTI is architecture-agnostic at the head; we instantiate it with three GCN architectures whose inductive biases differ significantly. **Single-level GCN** is a two-layer Kipf-Welling GCN at a single, relatively coarse level only (here L3). **Multi-scale concat** runs an independent two-layer GCN at two levels (here L2 and L3) in parallel and concatenates global-pooled embeddings before the head. **Hierarchical Graph U-Net** replaces dynamic pooling with the fixed index hierarchy above (from L2 to L3 to L4): a GCN block per encoder level, scatter-unpool with skip connections in the decoder, and global mean+max pool of the finest decoder level into the output head. Best models are selected on inner-fold validation AUC (or negative-MAE for regression).

2.4 Attribution stack

For each architecture we compute three complementary attributions: out-of-fold permutation importance [8] for tracts, per-node positions, covariates, and (in regression) per-target; integrated gradients on the input feature tensor [7]; and a sparse GNNExplainer mask [6] per held-out subject. The permutation importance metric is the drop in held-out AUC (or rise in held-out MAE for regression) when a feature group is shuffled across subjects in the outer-CV test fold. Crucially this is computed inside the nested-CV loop, so the model used for attribution has never seen the test subjects.

3 Application: Infant Down Syndrome Study

3.1 Dataset and cross-validation protocol

We evaluate on an infant Down syndrome (DS) cohort [14] with imaging at 6 months and outcome assessment at 24 months with 59 infants with DS and 68 control infants. Five diffusion metrics are assessed: FA, RD (Radial Diffusivity), AD (Axial Diffusivity), NDI, and ODI (Orientation Dispersion Index). For the prospective regression we predict scores from a behavioral assessment measured at 24 months of age, specifically the Vineland-III Daily Living Skills.

All experiments use a stratified 5-fold cross-validation repeated 3 times with inner-fold model selection.

3.2 Imputation evaluation

We evaluated the INR imputation on a held-out subset of 34 subjects with complete data by removing 7 random tracts per subject and reconstructing them from the rest. Normalized mean absolute error (NMAE) was lowest for INR/SIREN (0.520), followed by kNN-10 (0.576, +11% vs INR), kNN-5 (+16%), linear (+30%), median (+34%), and mean (+37%). The INR/SIREN imputation outperforms all baselines significantly ($p < 0.001$).

Table 1. Down syndrome classification: held-out CV AUC and F1 with covariates concatenated at the head. Baselines compute per-tract DTI/NODDI means; CONFETTI uses per-node features under the multi-resolution graph pipeline. Numbers are mean $\pm$ std over 15 outer folds (5×3 repeats), best dropout per model.

Method	AUC	F1
LogReg (per-tract means)	1.000 ± 0.000	0.994 ± 0.015
XGBoost (per-tract means)	0.983 ± 0.018	0.921 ± 0.052
CONFETTI - Single-level GCN (L3)	0.913 ± 0.125	0.809 ± 0.250
CONFETTI - Multi-scale concat (L2+L3)	0.931 ± 0.084	0.820 ± 0.128
CONFETTI - Graph U-Net (L2-L4)	0.940 ± 0.106	0.888 ± 0.093

3.3 Classification: CONFETTI is close to baselines, but with different attribution

Table 1 summarizes the held-out classification performance. Logistic regression on per-tract means saturates at a near perfect performance $AUC = 1.000$, $F1 = 0.994$; XGBoost reaches $0.983/0.921$. All three CONFETTI models land a bit lower, between 0.913 and 0.940 , with the Graph U-Net at the upper end. The gap is relatively small, likely due to the larger complexity of the GCN models.

Per-tract attribution tells a different story for the baselines and the GCN models. Figure 2 compares the inverse rank (highest bar = top-ranked tract) of the top ranking tracts. The two baselines put the *Mid Cerebellar Pontine (MCP) tract* at rank 1, with the *Corpus Callosum (CC) Lateral Genu tract* a close second. In contrast, all three CONFETTI permutation rankings put the *Anterior Commissure (AC) olfactory* at rank 1 by a wide margin, while still flagging the MCP tract as a secondary contributor. Furthermore, spearman rank correlation across all 66 tracts is 0.64 between the two baselines, but only (at most) 0.34 between any CONFETTI instantiation and any baseline.

3.4 Validating CONFETTI’s divergent finding

We ran five experiments to establish whether CONFETTI’s AC olfactory tract importance is a biological signal or a methodological artifact.

(1,2) *Raw biology and per-property channels.* Two-sample t -tests at each of the 52 AC olfactory nodes per property (260 tests, FDR-corrected [17]) yield $174/260$ significant at $q < 0.05$, and all 52 nodes are significant for at least one property; peak per-node Cohen’s d reaches $+1.10$ for AD and $+1.09$ for FA (Fig. 3, left). While the overall direction ($AD \uparrow$, $RD \uparrow$, $NDI \downarrow$, $ODI \uparrow$) matches the hypomyelination + reduced-neurite-density signature reported in DS cohorts [14, 16, 15], the group differences invert direction for FA, ODI, and NDI along the tract. This effect inversion cannot be captured by tract means and is likely one of the reasons the baselines do not rank the AC olfactory at the top.

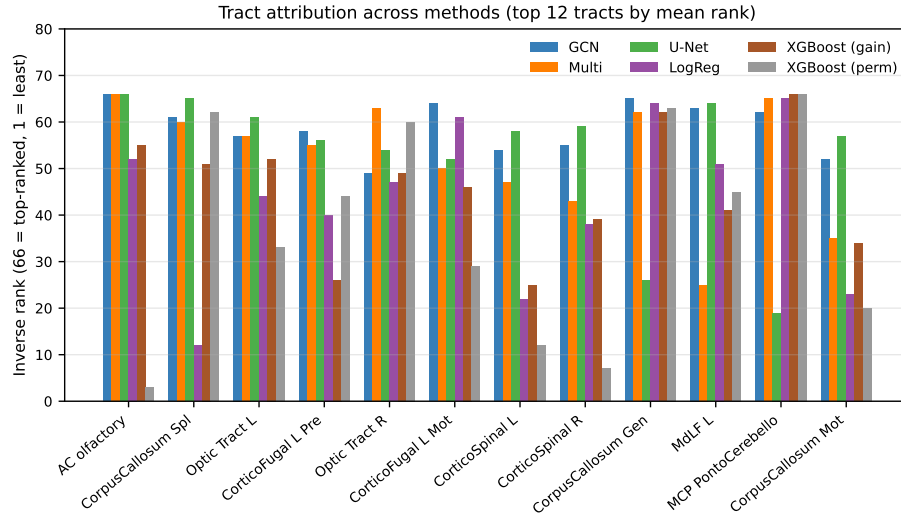


Fig. 2. Cross-method tract attribution. Per-method inverse rank (66 minus rank) for the top tracts. The three CONFETTI models agree on the AC olfactory at rank 1; the baselines rank the MCP and CC Lateral Genu tracts highest. The MCP and CC Lateral Genu tracts appear in $\geq 5/6$ top-10 lists.

Restricting the OOF permutation to AC olfactory and shuffling one property at a time isolates AD (+0.048 gcn, +0.036 unet) and RD as the predictive drivers; NDI and ODI contribute near zero. CONFETTI’s attribution at the AC olfactory tract therefore loads on the axial and radial diffusivity channels.

(3,4) *Necessity and sufficiency.* Replacing AC olfactory’s per-node values with the training-fold mean does not hurt classification (CV AUC 0.913 $\rightarrow$ 0.959 gcn, 0.931 $\rightarrow$ 0.929 multi, 0.940 $\rightarrow$ 0.944 unet): the models redistribute information across the remaining 65 tracts, consistent with the expected brain-wide DS pathology. Training on AC olfactory alone reaches 0.761/0.774/0.752 (gcn/multi/unet), well above chance from the ≤ 52 axis points (Fig. 3, right). AC olfactory is therefore informative but redundant for classification.

(5) *Within-tract spatial peak.* Aggregating the OOF GNNExplainer per-DS-subject masks places the spatial peak at arclength +13 mm across all three CONFETTI architectures, with $\geq 1.6\times$ activation versus other-tract nodes. This convergence under a different attribution mechanism than (1-4) indicates a likely stable spatial signal rather than a permutation artifact.

3.5 Predictive regression: 6-month imaging predicts later outcomes

We applied CONFETTI to a regression of the 24-month Vineland Daily Living Skills standardized scores. The performance of the regression is quite strong.

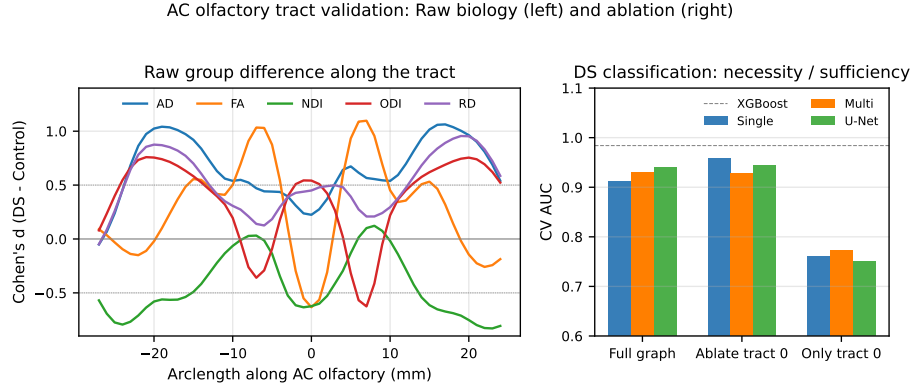


Fig. 3. Validation of the AC olfactory finding. Left: per-node Cohen's d along AC olfactory for each diffusion property; horizontal dotted lines mark $|d| = 0.5$. Right: classification AUC with the full graph, AC olfactory ablated, and only AC olfactory retained; dashed line is XGBoost.

Table 2. 6 month imaging predicting 24 month outcomes. Pearson r on held-out folds (mean over 3x5 folds; std in parentheses). **Bold** = best of row.

V24 outcome	Ridge baseline	XGBoost baseline	Graph U-Net
Vineland Daily Living Skills	0.32 (.19)	0.48 (.12)	0.51 (.13)

Table 2 shows held-out Pearson r for the Graph U-Net instantiation (the strongest CONFETTI head on this task) against the two baselines. CONFETTI outperforms both baselines.

Per-target attribution. Figure 4 shows CONFETTI's OOF tract permutation importance for the Graph U-Net head. AC olfactory is the top-attributed tract for the prediction, with a $\sim 4\times$ gap over the next-largest. The fold-wise std is wide (0.72), reflecting that the effect is very concentrated in some folds and modest in others — but the mean remains clearly the largest of any tract. The left optic tract is the strongest secondary contributor.

The same OOF protocol applied to the three covariates shows that sex has a predictive signal for the Daily Living score (+0.212), roughly half the AC olfactory tract effect. Contributions from gestational age at birth are much smaller and from DWI artifacts essentially zero.

4 Discussion

Under matched out-of-fold cross-validation, CONFETTI's top-ranked tract differs from that of the per-tract-mean baselines: CONFETTI puts the AC olfactory tract at rank 1, while the tract mean methods put the middle cerebellar peduncle at rank

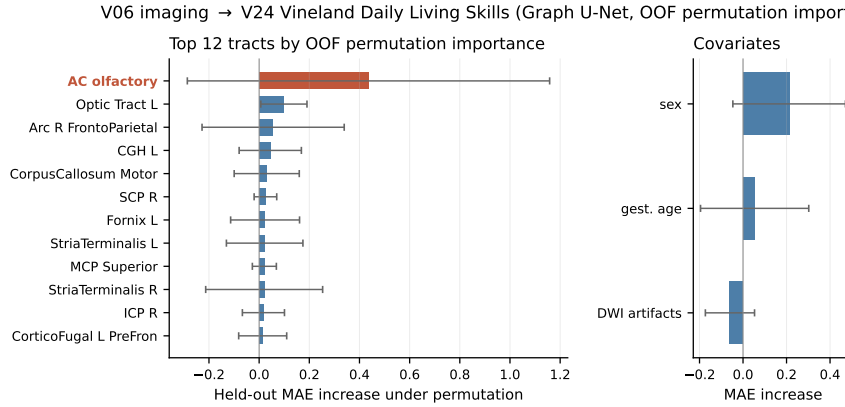


Fig. 4. CONFETTI per-target tract attribution for the 6-month diffusion → 24-month behavior predictive regression (Graph U-Net).

1 with the AC olfactory outside of the top-10. The follow-up experiments establish this as biological signal rather than artifact, and the same tract dominates the predictive outcome regression. The likely reasons the baselines miss this finding: (1) group differences invert direction along the tract which cannot be captured by tract means, (2) the AC olfactory has also particularly sharp group differences that mean-pooling averages away. Other architecture-independent consensus tracts remain visible in CONFETTI’s attribution.

The AC olfactory tract is generally understudied. Almost no single diffusion MRI study quantifies the AC olfactory on its own. When the AC is studied, quantification is limited to the AC as one bundle or to the larger posterior part. The results here show that particularly for DS studies, the AC olfactory tract should be included. This diffusion tract finding is supported by prior literature where severe, broad-based olfactory impairment affecting threshold detection, odor discrimination, and odor identification are observed in DS [19, 18].

It is worthwhile to note that the left and right optic tract are consistent secondary contributors. These tracts show an opposite directional signature to the AC olfactory tract: higher FA and lower RD/ODI indicating a seemingly more coherent microstructure pattern in DS, which is a biologically surprising finding. On the other hand, in a neurodevelopmental context, this pattern of diffusion MRI differences could also be interpreted as atypical development due to sparser fiber populations or thinner axons.

5 Conclusion

We introduced CONFETTI, a white-matter analysis framework that treats the whole tractography as a single off-grid geometric object rather than a set of per-tract means. Its methodological contributions are (i) a hierarchical multi-

resolution neighborhood graph over parametrized fiber axes preserving within-tract and between-tract structure in one shared hierarchy, (ii) a per-subject INR imputation for missing diffusion profiles that outperforms kNN, linear, mean and median baselines, and (iii) an architecture-agnostic GCN back-end with a nested-CV out-of-fold attribution stack. Together these components enable joint cross-tract attribution that recovers spatial signals per-tract-mean baselines do not reveal, illustrated here by the infant DS AC olfactory finding.

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